

allocation, and it cannot easily be compared to a total-return investing strategy with a static allocation. Whether time segmentation is a better strategy depends on the interactions of three issues:

- Because asset allocation is allowed to fluctuate, a time segmentation approach can have a very different asset allocation path than a total-returns investing approach. Is this acceptable to the retiree who has always been told that a more static (or at least a slowly changing) allocation is the right way to invest?
- Time segmentation approaches introduce varying degrees of effort in attempting to avoid selling stocks at inopportune times.
- By holding individual bonds to maturity, fixed-income assets do not need to be sold at a loss to support retirement income.

Exhibit 7.21 provides the ongoing probabilities of success for seven different strategies in supporting the retirement spending goal (spend \$40,000 initially with 2 percent cost-of-living adjustments in subsequent years) for up to forty years of retirement. The analysis is based on ten thousand Monte Carlo simulations starting from today's low bond yields with the capacity for rates to increase over time (as described in the technical appendix to chapter 3). Three of the strategies are the time segmentation rolling ladder approaches previously described with bond ladder lengths targeted at ten years. Ladders are extended either automatically, in years after positive market returns, or when remaining wealth exceeds the critical path level for that year of retirement.

The other four strategies are for total-returns investment portfolios. The first of these maintains a fixed 60.8 percent stock allocation throughout retirement, which matches the initial stock allocation after constructing a ten-year bond ladder at the start of retirement. This strategy is worthwhile to include because retirees using total-returns portfolios are more likely to maintain a relatively stable asset allocation in retirement. However, this strategy is not directly comparable to the time segmentation strategies because the asset allocation will be so different in subsequent years. The other three total-returns strategies use dynamic asset allocations designed to match the asset allocation generated by the time segmentation strategies for each Monte Carlo simulation in each year of retirement. Though real retirees would be unlikely to use such dynamic asset allocation strategies, since they